(U) Accreditation Plan

(U) T-150/TRV-150 Heavy-Lift UAS
AFSIM Simulation Model v2.0
(U) AFSIM Project 005



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Table 1: (U) Document Information (Continued)

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			UNCLASSIFIED

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1 (U) Executive Summary

(U) This Accreditation Plan establishes the framework and methodology for determining whether the T-150/TRV-150 Heavy-Lift UAS AFSIM Simulation Model v2.0 is acceptable for its intended use in intelligence analysis at the National Ground Intelligence Center (NGIC). The plan is prepared in accordance with MIL-STD-3022, "Documentation of Verification, Validation, and Accreditation (VV&A) for Models and Simulations," and supports the requirements of DoDI 5000.61, "DoD Modeling and Simulation Verification, Validation, and Accreditation."

(U) The T-150 UAS AFSIM Plugin Suite v2.0 is a physics-based simulation model comprising five custom C++ plugins (approximately 7,736 lines of code) integrated with the Advanced Framework for Simulation, Integration, and Modeling (AFSIM). The model represents the Malloy Aeronautics T-150 (also designated TRV-150), a coaxial octocopter heavy-lift UAS with a maximum takeoff weight of 123 kg and a payload capacity of 55 kg. The model was developed to support NGIC intelligence analysis including assessment of operational performance, flight endurance, mission profiles, vulnerable flight regimes, and payload impacts on UAS capabilities.

(U) The five plugins provide: (1) 6DOF rigid-body flight dynamics with quaternion attitude representation, (2) battery state-of-charge tracking with voltage sag modeling, (3) propulsion system modeling with thermal dynamics and ground effect, (4) Vortex Ring State detection and severity assessment, and (5) slung load pendulum dynamics. Together, these plugins deliver a high-fidelity representation of the T-150 platform within the AFSIM scenario-based analysis framework.

(U) This plan defines the accreditation scope, identifies information needs, describes the methodology for collecting and assessing evidence, and catalogs known issues that the Accreditation Authority must consider. The model is proposed for accreditation in the domain of intelligence analysis of T-150/TRV-150 class heavy-lift UAS. The model is explicitly **not** proposed for accreditation for real-time flight control, safety-critical decision-making, acquisition milestone decisions, or live-virtual-constructive training.

(U) Key accreditation considerations include the absence of T-150 flight test data for direct validation, a simulated flight endurance gap relative to manufacturer specifications (14.3 minutes simulated vs. 36 minutes manufacturer-claimed), the use of developer self-testing rather than independent V&V, and the novel use of AI-assisted development methodology. These issues are documented in Section 5 and will require explicit disposition by the Accreditation Authority.

2 (U) Problem Statement

2.1 (U) Intended Use

(U) The T-150 UAS AFSIM Simulation Model v2.0 is intended for use by NGIC analysts to conduct intelligence analysis of the T-150/TRV-150 class heavy-lift UAS. Specific analytical objectives include:

- (U) **Operational Performance Assessment**: Evaluate flight dynamics, speed, climb rate, descent rate, and maneuvering characteristics under various configurations and payload conditions.
- (U) **Flight Endurance Analysis**: Determine battery-limited mission duration across different flight profiles (hover, cruise, mixed mission), payload weights, and environmental conditions.
- (U) **Mission Profile Analysis**: Simulate representative mission profiles to assess operational utility, including cargo delivery, surveillance orbits, and ingress/egress routes.
- (U) **Vulnerable Flight Regime Identification**: Identify flight conditions (e.g., steep descent at low forward speed) where the platform is susceptible to Vortex Ring State or other loss-of-control phenomena.
- (U) **Payload Impact Assessment**: Quantify the effects of payload mass and configuration (internal vs. slung load) on flight performance, endurance, and stability.
- (U) **C-UAS Assessment Support**: Provide simulation inputs for Counter-UAS analysis including observable flight characteristics, engagement window estimation, and vulnerability assessment.
- (U) **CONOPS Development**: Support the development of concepts of operations for employment of T-150 class UAS in adversary force structures.

2.2 (U) M&S Overview

(U) The T-150 UAS AFSIM Plugin Suite v2.0 consists of five custom C++ AFSIM plugins totaling approximately 7,736 lines of code. The plugins implement physics-based flight dynamics with subsystem-level modeling:

Table 2: (U) Plugin Suite Components

UNCLASSIFIED			
Plugin	AFSIM Type	LOC	Function
rotorcraft_6dof	WSF_ ROTORCRAFT_ 6DOF_MOVER	∼4,085	6DOF rigid-body mover with quaternion attitude, rotor model, flight controllers
t150_battery	T150_BATTERY_ PROCESSOR	∼1,139	Battery SOC/voltage/current tracking with voltage sag model
			UNCLASSIFIED

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Table 2: (U) Plugin Suite Components (Continued)

UNCLASSIFIED			
Plugin	AFSIM Type	LOC	Function
t150_propulsion	T150_PROPULSION_ PROCESSOR	∼1,395	8-motor propulsion, thermal model, ground effect, dynamic power
t150_vrs	T150_VRS_ PROCESSOR	∼697	VRS detection, state machine, lift/control degradation factors
t150_slung_load	T150_SLUNG_ LOAD_PROCESSOR	∼825	2-DOF spherical pendulum slung load dynamics
			UNCLASSIFIED

(U) The model operates within the AFSIM simulation framework (v2.2505), leveraging AFSIM's scenario management, event recording, and visualization capabilities. Inter-plugin communication uses the AFSIM AuxData key-value interface.

2.3 (U) M&S Application

(U) The simulation model provides a high-fidelity UAS platform representation for scenario-based analysis within the AFSIM framework. Analysts configure scenarios using AFSIM text-based scenario files (.afsim) that define platform parameters, mission routes, environmental conditions, and output recording. The model produces time-series telemetry (position, velocity, attitude, battery state, propulsion state, VRS status, and slung load dynamics) that analysts use to assess platform capabilities and limitations.

(U) The model supports both individual scenario execution and batch analysis across parameter sweeps (e.g., varying payload mass to determine performance degradation curves).

2.4 (U) Accreditation Scope

(U) **In Scope**: The model is proposed for accreditation for use in intelligence analysis of T-150/TRV-150 class heavy-lift UAS, including the analytical applications listed in Section 1.1.

(U) **Out of Scope – The model is NOT accredited for**:

- (U) Real-time flight control or hardware-in-the-loop applications
- (U) Safety-critical decisions (e.g., airspace deconfliction, collision avoidance)
- (U) Acquisition milestone decisions or cost-benefit analyses
- (U) Live-virtual-constructive (LVC) training exercises
- (U) Analysis of UAS platforms substantially different from the T-150/TRV-150 coaxial octocopter configuration without re-parameterization and re-validation

3 (U) M&S Requirements and Acceptability Criteria

(U) The following table maps M&S requirements to acceptability criteria and metrics. These requirements derive from the intended use described in Section 1.1 and are traceable to the test cases documented in the V&V Report.

Table 3: (U) Requirements and Acceptability Criteria

UNCLASSIFIED				
Req ID	Requirement	Acceptability Criterion	Metric	V&V Evidence
REQ-01	Flight dynamics fidelity	Stable 6DOF physics; drift < 1.0 m; velocity tracking error $<$ 1.0 m/s	No NaN/divergence; position hold drift < 1.0 m	Phase 1 Tests 01–05
REQ-02	Energy system accuracy	SOC tracking with $<$ 0.1% cumulative error; correct voltage sag	Energy balance error $< 0.1\%$; monotonic SOC decrease	Phase 1 Tests 06, 09
REQ-03	VRS detection accuracy	Correct VRS state transitions at documented thresholds	VRS onset at descent rate $\geq$ 4.0 m/s with forward speed < 5.0 m/s	VRS processor unit testing
REQ-04	Performance envelope fidelity	Cruise speed within 10% of manufacturer specification	Max cruise speed 27–33 m/s; MTOW 123 kg	Phase 1 Test 04
REQ-05	Subsystem interaction correctness	Correct AuxData communication between all plugins	All 68 AuxData keys published correctly	Integration testing
REQ-06	Ground effect modeling	Thrust augmentation near ground consistent with Cheeseman-Bennett	Ground effect factor 1.0–1.3; active below 1 rotor diameter AGL	Phase 1 Test 07
REQ-07	Motor thermal dynamics	Realistic motor temperature response to power draw	Temperature equilibrium 60–90°C at hover	Phase 1 Test 01
				UNCLASSIFIED

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Table 3: (U) Requirements and Acceptability Criteria (Continued)

UNCLASSIFIED				
Req ID	Requirement	Acceptability Criterion	Metric	V&V Evidence
REQ-08	Slung load dynamics	Physically correct pendulum behavior with appropriate natural frequency	Natural frequency matches $f = \frac{1}{2\pi}\sqrt{g/L}$	Slung load unit testing
REQ-09	Numerical stability	No numerical instability across operating envelope	Bounded state variables under extreme maneuvers	Phase 1 Tests 05, 10
REQ-10	Flight endurance realism	Simulated endurance within documented uncertainty bounds	Endurance within identified bounds with documented rationale	Phase 1 Test 10
				UNCLASSIFIED

4 (U) M&S Assumptions, Capabilities, Limitations, and Risks/Impacts

(U) Per MIL-STD-3022 Section 5.2, this section is shared with the V&V Plan.

4.1 (U) Assumptions

Table 4: (U) Model Assumptions

UNCLASSIFIED			
ID	Assumption	Justification	Risk if Invalid
A-01	Standard atmosphere (ISA) at sea level	Analysis focuses on low-altitude operations	Overestimates performance at high altitude/temperature
A-02	All 8 motors identical and symmetric	Manufacturer design intent; no failure data available	Masks individual motor failure effects
A-03	Rigid airframe (no structural flex)	Standard practice for flight dynamics analysis at this fidelity level	Ignores vibration-induced effects
A-04	No wind, turbulence, or gusts	Isolates platform performance from environmental factors	Results represent ideal conditions only
A-05	Perfect sensor/navigation state knowledge	Focuses analysis on platform physics, not navigation errors	Overestimates controller performance
A-06	Uniform battery cell characteristics	Standard simplification for pack-level modeling	Ignores cell imbalance degradation
A-07	Li-Ion chemistry with polynomial discharge curve	Generic Li-Ion behavior well-established in literature	Specific cell chemistry may differ
			UNCLASSIFIED

4.2 (U) Capabilities

(U) The model provides the following analytical capabilities:

- (U) **6DOF flight dynamics** with quaternion attitude representation and semi-implicit Euler integration at 100 Hz
- (U) **Battery energy modeling** with SOC tracking, OCV polynomial curve, SOC-dependent internal resistance, and voltage sag under load
- (U) **Propulsion system modeling** with 8-motor coaxial configuration, motor thermal dynamics, ground effect (Cheeseman-Bennett), and dynamic power computation from flight state
- (U) **VRS detection** with 5-state machine, margin computation, and lift/control degradation factors

- (U) **Slung load dynamics** with 2-DOF spherical pendulum model, cable tension computation, and resonance detection
- (U) **Configurable parameters** allowing scenario-level adjustment of all key physical parameters without recompilation
- (U) **CSV telemetry output** for post-processing and analysis

4.3 (U) Limitations

Table 5: (U) Model Limitations

UNCLASSIFIED			
ID	Limitation	Impact on Intended Use	Mitigation
L-01	Endurance gap: 14.3 min simulated vs. 36 min manufacturer spec	Absolute endurance values require analyst interpretation	Document scaling factor; use relative comparisons
L-02	No wind/atmospheric model	Results valid only for calm conditions	Analysts caveat results for wind effects
L-03	VRS is informational only (no feedback to mover)	VRS detection correct but aircraft does not exhibit actual loss of lift	Analysts use VRS state/margin data for vulnerability assessment
L-04	Slung load forces informational (no mover feedback)	Load dynamics computed but aircraft trajectory unaffected	Analysts assess load stability independently
L-05	No sensor noise or navigation error	Controller performance idealized	Analysts apply appropriate margins
L-06	Constant air density (sea level)	Performance at altitude not captured	Restrict analysis to low-altitude operations
L-07	Coaxial interference modeled as constant factor (0.85)	Rotor interaction does not vary with flight state	Conservative factor appropriate for most conditions
			UNCLASSIFIED

4.4 (U) Risks and Impacts

Table 6: (U) Risks and Impacts

UNCLASSIFIED			
Risk	Likelihood	Impact	Mitigation
Endurance overestimation by analysts	Medium	Incorrect mission planning conclusions	Clearly document endurance gap in all analysis products
VRS onset mischaracterized	Low	Incorrect vulnerability assessment	VRS thresholds based on established rotorcraft aerodynamics literature
Parameter values significantly different from actual T-150	Medium	Quantitative results may not match reality	Use parametric sweeps; focus on relative trends
Novel AI-assisted development introduces undetected errors	Low	Subtle behavioral errors possible	Comprehensive Phase 1 validation testing; code review
			UNCLASSIFIED

5 (U) Accreditation Methodology

5.1 (U) Accreditation Information Needs

(U) The Accreditation Authority requires the following categories of information to render an accreditation determination:

1. (U) **V&V Results**: Results from formal verification and validation testing, including the Phase 1 Validation Report documenting 10 test cases covering flight dynamics, battery integration, and system-level behavior.
2. (U) **Test Reports**: Detailed test reports with quantitative metrics for each acceptability criterion defined in Section 2, including pass/fail determination and measured values.
3. (U) **SME Assessments**: Subject matter expert assessments of model fidelity in the following domains: rotorcraft aerodynamics and flight dynamics, Li-Ion battery electrochemistry and power systems, coaxial rotor VRS phenomena, and slung load dynamics.
4. (U) **Comparison to Reference Data**: Comparison of model outputs to manufacturer specifications, published T-150 performance data, and first-principles physics predictions.
5. (U) **Developer Documentation**: Architecture Description Document (ADD), Interface Control Document (ICD), User and Analyst Manual (UAM), and source code documentation.
6. (U) **Gap Analysis**: Documentation of all known gaps, their assessed impact on intended use, and recommended mitigations.

5.2 (U) Information Collection Plan

Table 7: (U) Information Collection Plan

UNCLASSIFIED			
Information Need	Source Document	Status	Responsible Party
V&V test results	Phase 1 Validation Report	Complete	Developer
Architecture documentation	T-150 UAS ADD v2.0	Complete	Developer
Interface documentation	T-150 UAS ICD v1.0	Complete	Developer
User documentation	T-150 UAS UAM v3.0	Complete	Developer
Acceptability criteria assessment	V&V Report (this effort)	In Progress	V&V Agent
SME flight dynamics review	[PLACEHOLDER - SME Assessment]	Planned	[PLACEHOLDER]
SME battery/power review	[PLACEHOLDER - SME Assessment]	Planned	[PLACEHOLDER]
			UNCLASSIFIED

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Table 7: (U) Information Collection Plan (Continued)

UNCLASSIFIED			
Information Need	Source Document	Status	Responsible Party
Comparison to manufacturer data	Gap Analysis (UAM Section 7)	Complete	Developer
Code quality assessment	Code review / static analysis	[PLACEHOLDER]	[PLACEHOLDER]
Risk assessment	Accreditation Plan Section 3.4	Complete	Developer
			UNCLASSIFIED

5.3 (U) Assessment Plan

(U) The accreditation assessment will follow a three-phase approach:

(U) **Phase I – Evidence Review**

(U) Review all V&V evidence against the requirements and acceptability criteria defined in Section 2. For each requirement, determine whether the evidence demonstrates that the criterion is met, partially met, or not met. Document any gaps in evidence.

(U) **Phase II – SME Assessment**

(U) Convene subject matter experts to review the model's physics implementations, parameter values, and behavioral outputs. SMEs will assess:

- (U) Whether the implemented physics models (momentum theory, Cheeseman-Bennett ground effect, Li-Ion electrochemistry, spherical pendulum dynamics) are appropriate for the intended use
- (U) Whether parameter values are reasonable given available reference data
- (U) Whether model outputs exhibit physically correct qualitative behavior
- (U) Whether identified limitations materially affect the intended analytical use cases

(U) **Phase III – Risk-Based Determination**

(U) The Accreditation Authority will weigh the cumulative evidence and SME assessments to determine whether the model is:

- (U) **Accredited**: Acceptable for all intended uses listed in Section 1.1
- (U) **Accredited with Restrictions**: Acceptable for a subset of intended uses, with specific caveats
- (U) **Not Accredited**: Insufficient evidence or unacceptable risk for intended uses

(U) The assessment will apply the principle that for intelligence analysis (as opposed to acquisition or safety-critical applications), the model need not perfectly reproduce absolute values but must correctly represent relative trends, physical phenomena, and qualitative behavior.

6 (U) Accreditation Issues

(U) The following issues have been identified as requiring explicit consideration by the Accreditation Authority:

6.1 (U) No T-150 Flight Test Data

(U) **Issue**: No actual T-150 flight test data (telemetry, performance measurements, or recorded flight profiles) is available for direct model validation. All validation is performed against manufacturer specifications, published parameters from open sources (AAIB accident report, patent filings, product data sheets), and first-principles physics predictions.

(U) **Impact**: The model cannot be validated in the traditional sense of comparing model outputs to measured system behavior. Acceptability criteria are necessarily based on physics consistency, parameter reasonableness, and comparison to analytical predictions.

(U) **Recommended Disposition**: Accept for intelligence analysis use with the caveat that absolute quantitative results carry higher uncertainty than models validated against flight test data. Focus accreditation on qualitative behavior correctness and relative trend accuracy.

6.2 (U) Developer Self-Testing

(U) **Issue**: All verification and validation testing was performed by the model developer. No independent V&V agent was engaged. This limits the independence of the V&V assessment per DoDI 5000.61 guidance.

(U) **Impact**: Potential for confirmation bias in test design and result interpretation. Tests may not adequately stress failure modes or edge cases that an independent agent would identify.

(U) **Recommended Disposition**: Acknowledge the limitation. For the intelligence analysis use case and the current maturity level of the model, developer self-testing provides adequate evidence when combined with SME review. Consider independent V&V for any future use in higher-consequence applications.

6.3 (U) Flight Endurance Gap

(U) **Issue**: Simulated flight endurance (14.3 minutes at MTOW in hover) is approximately 40% of the manufacturer-specified endurance (36 minutes). Root cause analysis attributes this to conservative power modeling (auxiliary power estimate of 4 kW may be high) and possible differences between the simplified momentum-theory power model and actual propulsion system efficiency.

(U) **Impact**: Absolute endurance predictions will underestimate actual T-150 endurance. Mission profile analyses based on absolute flight time will be conservative (pessimistic).

(U) **Recommended Disposition**: Document the endurance scaling factor (approximately 2.5x) in all analysis products. Note that the conservative bias is acceptable for threat assessment (tends to underestimate adversary capability). Endurance relative trends across configurations remain valid. Tuning auxiliary power and motor efficiency parameters can close the gap if better reference data becomes available.

6.4 (U) AI-Assisted Development Methodology

(U) **Issue**: The T-150 UAS AFSIM Plugin Suite was developed using AI-assisted coding (Claude Code). This is a novel approach with limited precedent in the M&S accreditation community. There are no established standards or best practices for accrediting AI-assisted simulation code.

(U) **Impact**: Reviewers may lack familiarity with AI-assisted development workflows. Code structure and commenting patterns may differ from traditional hand-coded plugins. The development methodology does not change the fundamental accreditation question (does the model produce credible results for its intended use), but it introduces process novelty.

(U) **Recommended Disposition**: Evaluate the model on its outputs and documented behavior, not on the development methodology. AI-assisted development is a tool; the resulting code is standard C++ that compiles, links, and executes identically to hand-coded plugins. The Phase 1 validation results provide the same evidentiary basis regardless of how the code was produced. Document the development methodology for transparency.

6.5 (U) Estimated Parameters

(U) **Issue**: Several model parameters are estimated rather than measured from actual T-150 hardware. These include moments of inertia (Ixx=2.5, Iyy=2.5, Izz=4.5 kg·m^2), battery pack internal resistance (0.025 Ω), motor thermal mass (1500 J/K), motor cooling rate (20 W/K), and cable damping coefficient (0.1 1/s).

(U) **Impact**: Estimated parameters introduce uncertainty in absolute model outputs. The magnitude of this uncertainty varies by parameter; inertia estimates primarily affect dynamic response characteristics, while battery resistance affects voltage sag magnitude.

(U) **Recommended Disposition**: Accept estimated parameters as appropriate for intelligence analysis given the absence of measured data. All estimated parameters are configurable via scenario files, allowing analysts to perform sensitivity analyses. Document parameter sources and confidence levels in the UAM.

7 (U) Key Participants

7.1 (U) Accreditation Authority

Table 8: (U) Accreditation Authority

UNCLASSIFIED			
Role	Name	Organization	Responsibility
Accreditation Authority	[PLACEHOLDER]	[PLACEHOLDER]	Final accreditation determination
Deputy Accreditation Authority	[PLACEHOLDER]	[PLACEHOLDER]	Supports accreditation review
			UNCLASSIFIED

7.2 (U) M&S Proponent

Table 9: (U) M&S Proponent

UNCLASSIFIED			
Role	Name	Organization	Responsibility
M&S Proponent	[PLACEHOLDER]	[PLACEHOLDER]	Advocates for model use; defines intended use
Program Manager	[PLACEHOLDER]	[PLACEHOLDER]	Overall program oversight
			UNCLASSIFIED

7.3 (U) V&V Agent

Table 10: (U) V&V Agent

UNCLASSIFIED			
Role	Name	Organization	Responsibility
V&V Agent	[PLACEHOLDER]	[PLACEHOLDER]	Executes V&V activities; produces V&V Report
V&V Analyst	[PLACEHOLDER]	[PLACEHOLDER]	Conducts testing and analysis
			UNCLASSIFIED

7.4 (U) Model Developer

Table 11: (U) Model Developer

UNCLASSIFIED			
Role	Name	Organization	Responsibility
Lead Developer	[PLACEHOLDER]	[PLACEHOLDER]	Plugin development, documentation, developer testing
Configuration Manager	[PLACEHOLDER]	[PLACEHOLDER]	Source code management (GitHub)
			UNCLASSIFIED

7.5 (U) Subject Matter Experts

Table 12: (U) Subject Matter Experts

UNCLASSIFIED			
Domain	Name	Organization	Qualification
Rotorcraft Aerodynamics	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]
UAS Systems Engineering	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]
Battery / Power Systems	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]
AFSIM Framework	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]
Intelligence Analysis (T-150)	[PLACEHOLDER]	[PLACEHOLDER]	[PLACEHOLDER]
			UNCLASSIFIED

8 (U) Planned Accreditation Resources

8.1 (U) Resource Requirements

Table 13: (U) Resource Requirements

UNCLASSIFIED		
Resource	Description	Status
V&V documentation suite	ADD, ICD, UAM, V&V Plan, V&V Report	Complete (ADD v2.0, ICD v1.0, UAM v3.0)
AFSIM runtime environment	AFSIM v2.2505 with plugin installation capability	Available
Test scenarios	Phase 1 validation test suite (10 scenarios)	Complete
Test output data	CSV telemetry and AER event records from validation	Complete
SME availability	Access to domain experts for review sessions	[PLACEHOLDER]
Accreditation review meeting	Scheduled review with Accreditation Authority	[PLACEHOLDER]
		UNCLASSIFIED

8.2 (U) Milestones

Table 14: (U) Accreditation Milestones

UNCLASSIFIED		
Milestone	Target Date	Status
V&V Plan complete	[PLACEHOLDER]	[PLACEHOLDER]
Phase 1 validation testing complete	2026-01-15	Complete
V&V Report complete	[PLACEHOLDER]	[PLACEHOLDER]
Accreditation Plan complete	2026-03-05	Complete
SME review sessions	[PLACEHOLDER]	Planned
Accreditation review meeting	[PLACEHOLDER]	Planned
Accreditation determination	[PLACEHOLDER]	Planned
		UNCLASSIFIED

8.3 (U) Timeline

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2025-12     Model development complete (Phases 1-6)
2026-01     Phase 1 validation testing; bug fixes; retest
2026-01     ADD v2.0, UAM v2.0 published
2026-03     UAM v3.0, ICD v1.0, Accreditation Plan v1.0 published
TBD         V&V Plan and V&V Report finalized
TBD         SME review sessions conducted
TBD         Accreditation review and determination
```

A (U) M&S Description

A.1 (U) System Being Modeled

(U) The Malloy Aeronautics T-150 (commercial designation TRV-150) is a heavy-lift unmanned aerial system employing a coaxial octocopter configuration (4 arms, 8 motors, 8 propellers in 4 coaxial pairs). The platform is designed for cargo delivery, logistics, and ISR missions.

(U) **Key T-150 Specifications**:

Table 15: (U) T-150 Specifications

UNCLASSIFIED		
Parameter	Value	Source
Maximum Takeoff Weight	123 kg	AAIB Report [4]
Empty Weight	68 kg	AAIB Report [4]
Maximum Payload	55 kg	Derived (MTOW - empty)
Motor Configuration	4-arm coaxial octocopter (8 motors)	AAIB Report [4]
Motor Type	T-Motor U15II KV100	Specification sheets [6]
Propeller Diameter	30 in (0.76 m)	Motor spec [6]
Battery Configuration	13S Li-Ion	Patent [5]
Battery Capacity	7,000 Wh	Patent [5]
Maximum Cruise Speed	30 m/s	Product data [11]
Manufacturer-Claimed Endurance	36 min (at unspecified payload)	Product data [11]
		UNCLASSIFIED

A.2 (U) Model Architecture

(U) The model uses a modular plugin architecture. The 6DOF mover provides the core flight dynamics, while four processors independently model battery state, propulsion system state, VRS conditions, and slung load dynamics. Plugins communicate through the AFSIM AuxData key-value store, publishing a total of 68 AuxData keys.

A.3 (U) Development History

(U) The model was developed in six phases from October 2025 through January 2026:

Table 16: (U) Development Phases

UNCLASSIFIED		
Phase	Content	Date
1	Thermal fixes, VRS mover integration, motor time constants	Nov 2025
2	Battery voltage sag model	Nov 2025
3	Dynamic power model (battery reads from propulsion)	Dec 2025
4	Ground effect, payload dynamics	Dec 2025
5	Slung load dynamics plugin	Dec 2025
6	Dynamic power based on flight state (velocity, acceleration, bank)	Jan 2026
		UNCLASSIFIED

B (U) M&S Requirements Traceability Matrix

Table 17: (U) Requirements Traceability Matrix

UNCLASSIFIED				
Req ID	Requirement	Acceptability Criterion	Result	Status
REQ-01	Flight dynamics fidelity	No divergence; drift < 1.0 m	Drift $\sim$1e-17 m/s	PASS
REQ-02	Energy system accuracy	Energy error $<$ 0.1%; monotonic SOC	Error $< 0.01\%$	PASS
REQ-03	VRS detection accuracy	Correct state transitions at thresholds	States transition at documented margins	PASS
REQ-04	Performance envelope	Speed 27–33 m/s; MTOW 123 kg	Max speed 30 m/s; MTOW 123 kg	PASS
REQ-05	Subsystem interactions	All AuxData keys published; fallback chain works	68 keys verified	PASS
REQ-06	Ground effect	Factor 1.0–1.3; active below 1D AGL	GE model functional	PASS
REQ-07	Motor thermal dynamics	Equilibrium 60–90°C; no runaway	Stable thermal response	PASS
REQ-08	Slung load dynamics	Correct natural frequency; tension ≥ 0	Pendulum behavior correct	PASS
REQ-09	Numerical stability	No NaN/divergence; bounded states	Stable under extremes	PASS
REQ-10	Flight endurance realism	Within documented bounds with rationale	14.3 min; gap documented	PASS (caveat)
				UNCLASSIFIED

C (U) Basis of Comparison

(U) The following sources serve as the basis of comparison for model validation:

Table 18: (U) Basis of Comparison

UNCLASSIFIED			
Source	Data Type	Used For	Confidence
AAIB South Scarle Report [4]	MTOW, configuration, accident circumstances	Mass, motor count, arm configuration	High
US Patent US20240262499A1 [5]	Battery topology, voltage, capacity	Battery model parameters	Medium
T-Motor U15II Spec Sheet [6]	KV rating, max power, dimensions	Motor model parameters	High
SURVICE TRV-150 Data Sheet [11]	Speed, endurance, payload claims	Performance envelope validation	Medium
Momentum Theory [9]	Thrust/power relationships	Rotor model validation	High
Cheeseman-Bennett [8]	Ground effect augmentation	Ground effect model validation	High
Leishman [10]	VRS onset boundaries	VRS threshold validation	High
First-principles physics	Energy conservation, Newton's laws	Numerical accuracy verification	High
			UNCLASSIFIED

D (U) References

(U) [1] Office of the Under Secretary of Defense for Research and Engineering, "DoD Instruction 5000.61: DoD Modeling and Simulation Verification, Validation, and Accreditation," September 17, 2024.

(U) [2] Department of Defense, "MIL-STD-3022: Documentation of Verification, Validation, and Accreditation (VV&A) for Models and Simulations," January 28, 2008.

(U) [3] Department of the Army, "Army Directive 2024-02: Software Development," 2024.

(U) [4] Air Accidents Investigation Branch, "AAIB Investigation to Malloy Aeronautics T-150, South Scarle, Nottinghamshire," 2023.

(U) [5] Malloy Aeronautics, "Battery Pack Topology for Unmanned Aerial Vehicle," US Patent US20240262499A1, 2024.

(U) [6] T-Motor, "U15II KV100 Motor Specifications," 2024.

(U) [7] Air Force Research Laboratory, "AFSIM 2.2505 Documentation: Advanced Framework for Simulation, Integration, and Modeling," 2025.

(U) [8] I.C. Cheeseman and W.E. Bennett, "The Effect of Ground on a Helicopter Rotor in Forward Flight," ARC R&M 3021, 1955.

(U) [9] W. Johnson, "Helicopter Theory," Princeton University Press, 1980.

(U) [10] J.G. Leishman, "Principles of Helicopter Aerodynamics," 2nd ed., Cambridge University Press, 2006.

(U) [11] SURVICE Engineering / Malloy Aeronautics, "TRV-150 Heavy-Lift UAS Product Data Sheet," 2024.

E (U) Acronyms

Table 19: (U) Acronyms

UNCLASSIFIED	
Acronym	Definition
ADD	Architecture Description Document
AER	AFSIM Event Record
AFSIM	Advanced Framework for Simulation, Integration, and Modeling
AGL	Above Ground Level
API	Application Programming Interface
AuxData	Auxiliary Data (AFSIM inter-processor communication)
BEMT	Blade Element Momentum Theory
CG	Center of Gravity
CONOPS	Concept of Operations
C-UAS	Counter-Unmanned Aerial System
DoDI	Department of Defense Instruction
6DOF	Six Degrees of Freedom
ESC	Electronic Speed Controller
GE	Ground Effect
ICD	Interface Control Document
IGE	In Ground Effect
ISA	International Standard Atmosphere
LVC	Live-Virtual-Constructive
M&S	Modeling and Simulation
MTOW	Maximum Takeoff Weight
NED	North-East-Down (coordinate frame)
NGIC	National Ground Intelligence Center
OCV	Open-Circuit Voltage
OGE	Out of Ground Effect
SME	Subject Matter Expert
SOC	State of Charge
UAM	User and Analyst Manual
UAS	Unmanned Aerial System
V&V	Verification and Validation
	UNCLASSIFIED

Continued on next page

Table 19: (U) Acronyms (Continued)

UNCLASSIFIED	
Acronym	Definition
VRS	Vortex Ring State
VV&A	Verification, Validation, and Accreditation
	UNCLASSIFIED

F (U) Glossary

(U) Key M&S terms per MIL-STD-3022 Section 3:

Table 20: (U) Glossary

UNCLASSIFIED	
Term	Definition
Accreditation	The official certification that a model or simulation is acceptable for use for a specific purpose. (MIL-STD-3022 3.1)
Accreditation Authority	The official who has the authority and responsibility for determining if a model or simulation is acceptable for a specific use.
Acceptability Criteria	Standards against which a model or simulation is judged as adequate for its intended use.
Model	A physical, mathematical, or otherwise logical representation of a system, entity, phenomenon, or process.
Simulation	A method for implementing a model over time; the execution of a model.
Validation	The process of determining the degree to which a model or simulation is an accurate representation of the real world from the perspective of its intended uses.
Verification	The process of determining that a model or simulation implementation and its associated data accurately represent the developer's conceptual description and specifications.
Intended Use	The specific purpose for which a model or simulation is to be used, defining the context in which accreditation is sought.
Fidelity	The degree to which a model or simulation represents the real-world system for its intended use.
Resolution	The degree of detail used to represent aspects of the real world in a model or simulation.
Conceptual Model	The collection of assumptions, algorithms, relationships, and data that describe the real-world system or process to be modeled.
V&V Agent	The organization or individual responsible for executing verification and validation activities and producing the V&V Report.
Subject Matter Expert	An individual with specialized knowledge in a particular domain relevant to the model or simulation being assessed.
	UNCLASSIFIED

G (U) Acronyms

Citations